

HARSH RAJ

 github.com/harshraj172  harsh777111raj@gmail.com  [linkedin.com/in/harsh-raj](https://www.linkedin.com/in/harsh-raj)  [Google Scholar](#)

EDUCATION

Northeastern University, Boston, MA
Master of Science in Computer Science

May 2027 (expected)
GPA: 3.6/4.0

Delhi Technological University, New Delhi, India
Bachelor of Technology in Engineering Physics

August 2023
GPA: 3.4/4.0

TECHNICAL SKILLS

Languages: Python, C++, CUDA, Java, SQL, HTML, JavaScript, Bash

Frameworks/Libraries: PyTorch, FastAPI, RayServe, RabbitMQ, MongoDB, DynamoDB, Redis, PyTest, VectorDBs

DevOps/Cloud: Docker, Kubernetes, Terraform, Azure, AWS, RunPod, Petals, CI/CD

Other: Git, Vim, SSH, Linux, RESTful APIs, Web-Sockets

Model Expertise: LLM Pre-training, LLM Post-training, Large Scale Data Synthesis

PROFESSIONAL EXPERIENCE

Research Assistant, Forschungszentrum Jülich
Data Curation and Scaling Laws

July 2025 – Present

- Contribute to EU Foundational Model projects ([OpenEuroLLM](#), [ELLIOT](#)) on JUPITER supercomputer
- Lead ablation studies on dataset mixtures and scaling laws on LLM training data at the petabyte scale
- **Impact:** Optimized multi-node data processing pipelines using Ray, achieving 5x speedup in deduplication, filtering and map reduction operations

Co-Founder, Ontocord AI (Open Source)
Pre-training Data Curation

Jan 2024 – Jan 2025

- Co-founded open-source initiative focused on creating high-quality permissive datasets for LLMs
- Developed data processing pipelines with strategies for mixing, formatting, upsampling & interleaving
- **Impact:** Published [MixtureVitae-300BT](#) achieving 10% performance improvement over existing permissive data sets

Applied Scientist, Vijil AI
LLM Red-Teaming

Feb 2024 – Nov 2024

- Contributed to the [vijil-evaluation](#) platform for red-teaming of enterprise AI agent workflows. Integrated more than 10 benchmark datasets for vulnerability assessment
- Developed the vijil-fuzzer for adversarial text mutation. It achieved a 85% attack rate against GPT-4o
- **Impact:** The vijil-evaluate service is used by enterprise clients including [DigitalOcean](#) and [Groq](#)

Machine Learning Engineer, Yield Inc
Post-training and Data Curation

Sept 2022 – Jan 2024

- Built synthetic data generation and finetuning infrastructure for LLM agents - [synchaev](#) and [mandrill](#)
- Contributed to [Cacti](#), a Web3 transaction chatbot that simplifies blockchain interactions through natural language
- **Impact:** To the best of our knowledge, it was the first open-source chatbot that used custom tool calling (we called it widgets) before any LLM had native tool calling capabilities

SELECTED PUBLICATIONS

Improving Consistency in Large Language Models through Chain of Guidance **1st Author**

TMLR 2025

Defences against Reverse Preference Attacks **3rd Author**

AAAI 2025

Measuring Reliability of Large Language Models through Semantic Consistency **1st Author**, *Best Paper* NeurIPS 2022

PROJECTS

terminal-bench (800+ GitHub ★) Language agent sandbox evaluation framework (**Co-author & Maintainer**)

inspect-ai (1300+ GitHub ★) Industry-standard LLM evaluation suite (**Contributor**)

AWARDS AND ACHIEVEMENTS

[SWISS AI](#) Grant Recipient – Awarded 200k GPU hours and CHF 61,000 in funding

2025

[Vision and Language Navigation](#) – Secured 3rd position (at the time of submission) on the R2R benchmark

2024

[Rank 7](#) – Codeforces Raif ML Round 1 Competition

2021